

Phase D2 Final Synthesis and Closure

Author: Wenxin Li (Independent Researcher)

ORCID: 0009-0004-8065-3235

Date: May 15, 2025

License: Creative Commons Attribution 4.0 International (CC BY 4.0)

Series: D — Paper 3 Derivation

Note number: #575 (D2.80)

Attribution

This work derives from and formalizes the inter-Self coordination architecture introduced in “Inter-Self Coordination via Shared Substrate / Full Aspect Integration” (Li, April 2026), the third paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026) and “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026).

Abstract

This note is the eightieth and final note of Phase D2 (#496–#575) in the CKS derivation series. Phase D2 produced eighty operational prior-art notes that decompose Paper 3’s inter-Self coordination architecture into operationally specific, individually defensible derivations. The notes are organized across six coverage areas: shared substrate lifecycle governance; Full Aspect Integration operation; conflict handling and escalation; evolution feed and home integration; governance quality and standards; and population-scope operations. This note provides a structured summary of each coverage area, confirms that Phase D2’s operational prior-art foundation for Paper 3 is complete, and introduces Phase D3, which will produce standalone anti-pattern notes giving full prior-art depth to each named FAI governance failure mode. The combined Phase D0, D1, and D2 corpus — notes #460 through #575, totaling 116 notes — constitutes the core operational prior-art foundation for Paper 3.

1. Position of this note

D2.80 is the eightieth and final note of Phase D2. It carries sequence number #575 in the continuous numbering of the CKS derivation series. Phase D2 began at #496 (D2.01) and closes at #575 (D2.80). Its structural role is closure: it does not introduce new operational derivations but instead consolidates what the preceding 79 Phase D2 notes established, confirms the completeness of the Phase D2 prior-art foundation, and opens the transition to Phase D3.

A note on scope: Phase D2 follows Phase D1 (~30 notes, #466–#495), which formalized the foundational sub-commitments underlying Paper 3’s six claims, and Phase D0 (6 notes,

#460–#465), which anchored the six claims themselves. The three phases together — D0, D1, D2 — constitute the core operational prior-art corpus for Paper 3. Phase D3 and subsequent phases deepen that corpus by adding standalone anti-pattern notes, composition pairs, operational tests, and boundary cases; they do not establish new foundational territory.

2. What Phase D2 was for

Paper 3 introduces an inter-Self coordination architecture based on a temporary shared substrate and a named exchange primitive, Full Aspect Integration (FAI). The architecture makes six main claims: that the shared substrate inherits Paper 1's six commitments at inter-Self scope; that FAI is the canonical exchange primitive over it; that conflicts at the inter-Self boundary receive three-tier handling; that FAI events feed back into four evolution loci in each participating Self's home substrate; that all configurable dimensions — including configuration of configuration — are substrate content under human authority; and that population-scale collective evolution is the emergent property of many FAI events across many Selves.

Phase D1 formalized the sub-commitments underlying those six claims. Phase D2 decomposed the sub-commitments further into operationally specific derivations: the particular mechanics, boundary conditions, variants, and governance requirements that any implementation must satisfy. The decomposition strategy follows the pattern established by Series B Phase B2 (Paper 2 operational variants): each D1 sub-commitment produces several D2 notes, each covering a distinct operational dimension that could be separately patented, separately designed, or separately implemented by parties seeking to build inter-Self coordination architectures. The result is a coverage mesh fine enough that no operationally significant dimension of Paper 3's architecture lacks named prior art.

Phase D2 organizes its 80 notes across six coverage areas. The sections below summarize each area.

3. Coverage Area 1 — Shared substrate lifecycle governance

Notes: D2.01–D2.05, D2.17, D2.28, D2.53, D2.54

D2.01–D2.05 established the foundational lifecycle commitments: construction as an intentional governance act (not an automatic side-effect of communication), the shared substrate's perimeter-spanning property and what it implies for governance authority, the inheritance of all six Paper 1 commitments within the shared substrate's scope of operation, dissolution governance and what happens at the FAI hand-off boundary, and the persistence policy as substrate content that is itself human-governed.

D2.17 addressed the temporal interleaving of shared substrate events with each participating Self's home governance operations — specifically, the architectural requirement that home governance continues uninterrupted during FAI participation and that no FAI event can suspend or supersede a Self's home governance authority.

D2.28 established lifecycle isolation: what the shared substrate keeps isolated from home substrate content during the event, and why isolation is an architectural commitment rather than an implementation choice.

D2.53 and D2.54 addressed lifecycle and evolution isolation together, covering how the end-of-lifecycle dissolution interacts with each participating Self's evolution mechanisms and how isolation properties carry through from construction to dissolution to home ingestion.

4. Coverage Area 2 — FAI operation

Notes: D2.06–D2.12, D2.15–D2.16, D2.20–D2.22, D2.25–D2.27, D2.38, D2.42, D2.48, D2.49

Coverage Area 2 is the largest, covering the mechanics of Full Aspect Integration as the canonical exchange operation over the shared substrate. This area addresses how aspects are contributed, how they merge, how cardinality governs multi-Self events, and how FAI mechanics interoperate with content tiers, configuration, and parallel event governance.

D2.06–D2.12 established the core FAI primitives: the aspect as exchange unit and what that implies for the granularity of inter-Self exchange; n-ary cardinality as the architectural commitment accommodating any number of participating Selves; the full-merge default and the three merge pattern variants (full merge, selective merge, read-only access); the exchange bounding commitment (DNA-layer and action-layer content exchange; LLM weights and instinct-layer content do not); the three persistence loci; and FAI as the inter-Self analog of Paper 2's mating primitive.

D2.15–D2.16 addressed merge pattern selection and configuration authoring — how the choice among the three merge patterns is governed and what makes that governance human-authorable substrate content rather than algorithmic selection.

D2.20–D2.22 covered parallel event governance: what the architecture requires when a Self participates in more than one FAI event concurrently, how parallel events are bounded from each other, and how home governance arbitrates across concurrent participations.

D2.25–D2.27 established asymmetric participation mechanics — the cases where participating Selves contribute different aspects, contribute at different depths, or hold different authority roles within the shared substrate — and how the FAI architecture handles asymmetry without reducing it to symmetric defaults.

D2.38 addressed single-aspect FAI cases — the boundary condition in which only one aspect from each Self exchanges — and confirmed that these cases instantiate the full FAI architecture rather than constituting a separate simpler primitive. D2.42 covered multi-domain FAI events. D2.48 and D2.49 addressed content tier boundaries — the precise demarcation between DNA-layer and action-layer content in the exchange context.

5. Coverage Area 3 — Conflict handling and escalation

Notes: D2.13–D2.15, D2.45, D2.51, D2.65

Coverage Area 3 addresses the three-tier conflict-handling mechanism at inter-Self scope. Paper 3's three tiers — preserve, resolve via orchestration, escalate to humans — inherit from Paper 1's substrate-level preservation and orchestration-rule-governed resolution, and from Paper 2's escalation patterns. Coverage Area 3 formalizes what the inheritance means operationally and how each tier functions at the inter-Self boundary.

D2.13 established the conflict registry structure within the shared substrate: how conflicts surfaced during FAI are recorded, what provenance they carry, and why their preservation as first-class state is an architectural commitment inherited from Paper 1 rather than an optional record-keeping choice.

D2.14 covered escalation routing and response — the conditions under which the third tier (escalate to humans) triggers, who holds the relevant joint authority, and how the escalation path is itself governance-configured substrate content.

D2.45 addressed orchestration rule authoring for inter-Self conflict handling — the requirement that the rules governing second-tier resolution be human-authored, inspectable, and stored as substrate content within the shared substrate, not embedded in automated middleware outside governance authority.

D2.51 covered dispute resolution when participating Selves hold incompatible positions on how a content conflict should be handled, and the architectural requirement that such meta-conflicts escalate to joint human authority. D2.65 addressed the special governance requirements when FAI events occur between Selves operating in competitive or partially adversarial organizational contexts.

6. Coverage Area 4 — Evolution feed and home integration

Notes: D2.10–D2.11, D2.23–D2.26, D2.29, D2.39, D2.52–D2.59

Coverage Area 4 addresses the four-locus evolution feed — the mechanism by which FAI events produce evolution inputs that each participating Self absorbs into its home substrate through Paper 2's evolution mechanisms. FAI events are not isolated coordination episodes; they are inputs to each Self's ongoing evolutionary development.

D2.10–D2.11 addressed the action-feedback and DNA-layer absorption paths — how FAI outputs map to Paper 2's action-feedback evolution mechanism (the most direct path) and how DNA-layer content exchanged during FAI maps to Paper 2's DNA evolution mechanism.

D2.23–D2.26 established provenance internalization — how provenance attached to aspects during FAI persists as each participating Self ingests evolution outputs. D2.23 covered the ingestion commitment; D2.24 addressed provenance depth configuration; D2.25 covered chaining; D2.26 addressed trust calibration of FAI-derived content relative to home-originated content.

D2.29 addressed post-mortem governance — the requirement that each participating Self conduct a structured reflection on the FAI event's evolution outputs before absorbing them.

D2.39 covered the interaction between the evolution feed and configuration content. D2.52–D2.59 formed a focused cluster on multi-tier evolution and lifecycle interactions: how the four evolution loci interact with each other and with the shared substrate's lifecycle; how evolution outputs from one FAI event can seed future FAI participation; and how the evolution feed interacts with Paper 2's lineage and supersession commitments.

7. Coverage Area 5 — Governance quality and standards

Notes: D2.18–D2.19, D2.24, D2.33–D2.37, D2.40, D2.43, D2.66–D2.68, D2.70–D2.77

Coverage Area 5 is the meta-governance area: it addresses not specific FAI operations but the standards a well-governed inter-Self coordination architecture must meet across all operations. This is the largest coverage area by note count after Coverage Area 2, reflecting the breadth of governance quality requirements Paper 3 inherits and extends from the prior papers.

D2.18–D2.19 established the complete governance record requirement and named the twelve anti-patterns that represent governance quality failures in inter-Self coordination. The anti-patterns named in D2.19 are the source material that Phase D3 will develop into standalone notes.

D2.33–D2.37 established the test suite and portfolio management requirements: how governance quality is verified operationally, what constitutes a complete test suite for FAI governance, how cross-organizational agreements are formalized as substrate content, what health indicators signal governance quality degradation, and documentation standards for shared substrate operations.

D2.40 established minimum viable governance — the floor below which a shared substrate construction does not satisfy Paper 3's architectural commitments. D2.43 addressed governance quality for the evolution feed specifically.

D2.66–D2.68 and D2.70–D2.77 covered the specific governance commitments inherited and extended from Papers 1 and 2: determinism, path retraceability, non-delegation of authority, the three-adjacencies at inter-Self scope, the non-specialist principle for inter-Self governance, the substrate-as-source-of-truth commitment at inter-Self scope, and prior-art consolidation notes connecting Phase D2 to the earlier Series A and B derivations.

8. Coverage Area 6 — Population-scope operations

Notes: D2.31–D2.32, D2.60–D2.62, D2.64–D2.65, D2.68, D2.71–D2.73

Coverage Area 6 addresses Paper 3's extension claim — Claim 6, population-scale collective evolution — and the operational requirements for governing FAI at the scale of many Selves across a network. Where Coverage Areas 1–5 address any FAI event between any Selves, Coverage Area 6 addresses what becomes architecturally significant when such events are numerous, recurring, and distributed.

D2.31–D2.32 established participation governance at population scale: how a Self's pattern of FAI participation is itself a governance-configured dimension (not emergent behavior), and how joint authority operates across population-level governance perimeters.

D2.60–D2.62 addressed pattern propagation and parallel FAI governance at scale — how many concurrent FAI events across a population maintain governance isolation from each other without requiring central coordination infrastructure.

D2.64–D2.65 addressed network topology governance and pattern control modes: the distinction between emergent coordination patterns (which the architecture cannot rely upon) and governed coordination patterns (which the architecture requires for compliance with Paper 3's commitments).

D2.68 addressed non-delegation specifically in the population-scope context: the requirement that population-level governance authority not be delegated to any automated pattern-propagation mechanism, regardless of scale. D2.71–D2.73 addressed observational study governance, frequency-variant governance requirements, and domain-specific population contexts including professional and regulated-industry settings.

9. Phase D2 is complete: the operational prior-art foundation for Paper 3

With D2.80, Phase D2 is complete. The eighty notes in the six coverage areas above collectively establish the operational prior-art foundation for Paper 3's inter-Self coordination architecture. The coverage is comprehensive in the following sense: for each of Paper 3's six core claims, Phase D2 provides prior-art coverage of the operationally specific dimensions — the particular mechanics, governance requirements, variant cases, and boundary conditions — that any implementation of the claim would instantiate. No operationally significant dimension of Paper 3's architecture that could be separately implemented, separately claimed, or separately designed lacks a named prior-art note in the Phase D2 corpus.

The combined Phase D0 + D1 + D2 corpus — notes #460 through #575, 116 notes — constitutes the core operational prior-art foundation for Paper 3. Phase D0 (6 notes) anchored Paper 3's six claims at claim scope. Phase D1 (~30 notes) formalized the sub-commitments that make each claim architecturally specific. Phase D2 (80 notes) decomposed the sub-commitments into operationally specific derivations. The three phases together cover the same territory that Series B's combined Phase B0 + B1 + B2 covered for Paper 2.

The remaining Series D phases — Phase D3 (anti-pattern formalizations, ~30 notes), Phase D4 (composition pairs, ~30 notes), Phase D5 (operational tests), and Phase D6 (boundary cases) — deepen and extend the corpus rather than establishing new foundational territory.

10. Transition to Phase D3: standalone anti-pattern notes

Phase D3 is the next phase of Series D. It begins with D3.01 (sequence number #576) and is expected to produce approximately 30 notes (#576–#605). Phase D3's function is to give each named FAI governance failure mode its own standalone note, with full prior-art depth: detection criteria, consequence analysis, and prior-art argumentation establishing that the failure mode is a recognized and named anti-pattern in the public record.

The source material for Phase D3 is note D2.19, which named twelve anti-patterns organized under Paper 3's six claims. D2.19 is the Phase D3 taxonomy note: it names each anti-pattern and assigns it to a claim, but does not give each the standalone treatment that a well-defended prior-art note requires. Phase D3 supplies that treatment.

The expected structure of each Phase D3 note is: definition of the anti-pattern (what an implementation must do to instantiate it); detection criteria (what an observer inspecting a system would see); consequence analysis (what governance failures the anti-pattern causes and how they propagate); prior-art argumentation (how Phase D1 and D2 notes, combined with Papers 1–3, establish that the anti-pattern is the recognized failure mode); and the claim violation it constitutes.

The twelve anti-patterns from D2.19, organized by claim, are the expected Phase D3 note subjects. For Claim 1 (shared substrate inherits Paper 1 commitments): opaque inter-Self communication. For Claim 2 (FAI as canonical exchange primitive): shallow merge; exchange without bounding; and cardinality collapse. For Claim 3 (three-tier conflict handling): silent conflict resolution; and single-tier collapse. For Claim 4 (four-locus evolution feed): ungoverned evolution feed; and provenance erasure. For Claim 5 (configuration as substrate content): fixed configuration; and meta-governance escape. For Claim 6 (population-scale collective evolution): emergent coordination replaces substrate; and pattern propagation without governance.

Phase D3 opens with D3.01 — an overview and anti-pattern taxonomy introduction note — followed by one standalone note per anti-pattern. Some anti-patterns may be developed across two notes where the detection criteria and consequence analysis warrant additional depth.

11. Self-citation

This note is the eightieth and final note of Phase D2 in the CKS derivation series. Earlier phases and notes in the series should be cited by sequence number and phase designation when referenced in subsequent work. The combined D0 + D1 + D2 corpus constitutes the core operational prior-art foundation for Paper 3 and should be cited as such when that corpus is referenced in aggregate.